QUASI-AFFINENESS PUSHES DOWN ALONG A FINITE SURJECTIVE MORPHISM IN TWO SITUATIONS

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 3915-3924 and proof span 3926-3932 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof is a case-by-case reduction in two short paragraphs, one per hypothesis; the author's enumeration is preserved.

Editorial note. 3 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A finite locally free morphism of degree d has a canonical norm of degree d, and the norm of a quasi-affine morphism admitting a norm is quasi-affine. Cited at proof lines 3926-3928, both statements being previous cards of this lane. Gives the first case by composing the norm construction with the quasi-affineness statement.
- The norm statement in the normal case, applicable after replacing the source by an irreducible component dominating the target. Cited at proof lines 3929-3932. Gives the second case.

Source reference. Original lemma 0BD2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14300–14300. Same-chapter reference used in the retained author text at source line(s) 3846, 3906, 3928. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BD3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14301–14301. Same-chapter reference used in the retained author text at source line(s) 3912, 3931. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's

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Source reference. Original lemma 0BD1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14299–14299. Same-chapter reference used in the retained author text at source line(s) 3928. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\pi : X \to Y$ be a finite surjective morphism of schemes. Assume that X is quasi-affine. If either*

(1) *π is finite locally free, or*
(2) *Y is an integral normal scheme*

then Y is quasi-affine.

Proof. Case (1) follows from a combination of Lemmas 0BD2 and 0BD1. In case (2) we first replace X by an irreducible component of X which dominates Y (viewed as a reduced closed subscheme of X). Then we can apply Lemma 0BD3. □